

Cybersecurity Internship – Week 1 Documentation

Security Improvements Implemented

1. Input Validation

The validator library was used to validate user email input during login and registration.

2. Password Hashing

Weak MD5 hashing was replaced with secure bcrypt hashing for better password protection.

1495	noman123@gmail.com	20e4080aee9060539d0b63958056ab38	<script>al...
1496	xss@test.com	827ccb0eea8a706c4c34a16891f84e7b	<script>al...
1497	secure@test.com	\$2b\$10\$8HJHcQKMK5AMRs8m9wj1pe...	Mian
	NULL	NULL	NULL

3. JWT Authentication

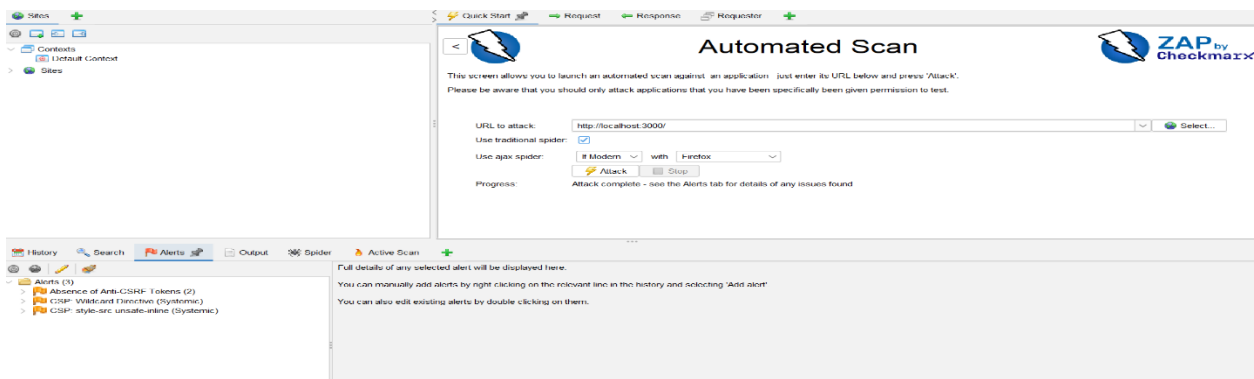
JWT token-based authentication was implemented after successful login.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\Cyber Security- Internship\project\User Management App> npm start
GET /favicon.ico 404 29.686 ms - 1472
GET /register 200 58.787 ms - 984
GET /stylesheets/style.css 304 2.476 ms - -
POST /register 302 335.885 ms - 56
GET /login 304 38.649 ms - -
GET /stylesheets/style.css 304 1.003 ms - -
JWT Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MTQ5NywiZW1hbmwiOiJzZW50cmVudGVzdC5jb20iLCJpYXQiOiE3Nzk2Mjg5MjUsImV4cCI6MTc3OTYzMTcyNX0.Ae26zihQIleMFLfgXfLEGVYsHjXAJ416KOPwrmCJ7o
POST /login 302 106.262 ms - 46
GET / 200 57.109 ms - 4102
GET /stylesheets/style.css 304 2.904 ms - -
```

4. Helmet.js

Helmet middleware was added to improve HTTP security headers and reduce vulnerabilities.



Testing Performed

- Invalid email testing using validator
- bcrypt password hash verification in database
- JWT token generation testing
- OWASP ZAP re-scan after security fixes

Results

- Passwords are now securely stored using bcrypt.
- Invalid email inputs are blocked.
- JWT authentication working successfully.
- OWASP ZAP showed significant reduction in security header vulnerabilities.